Exploratory data analysis

How to start making data talk

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Today's objectives

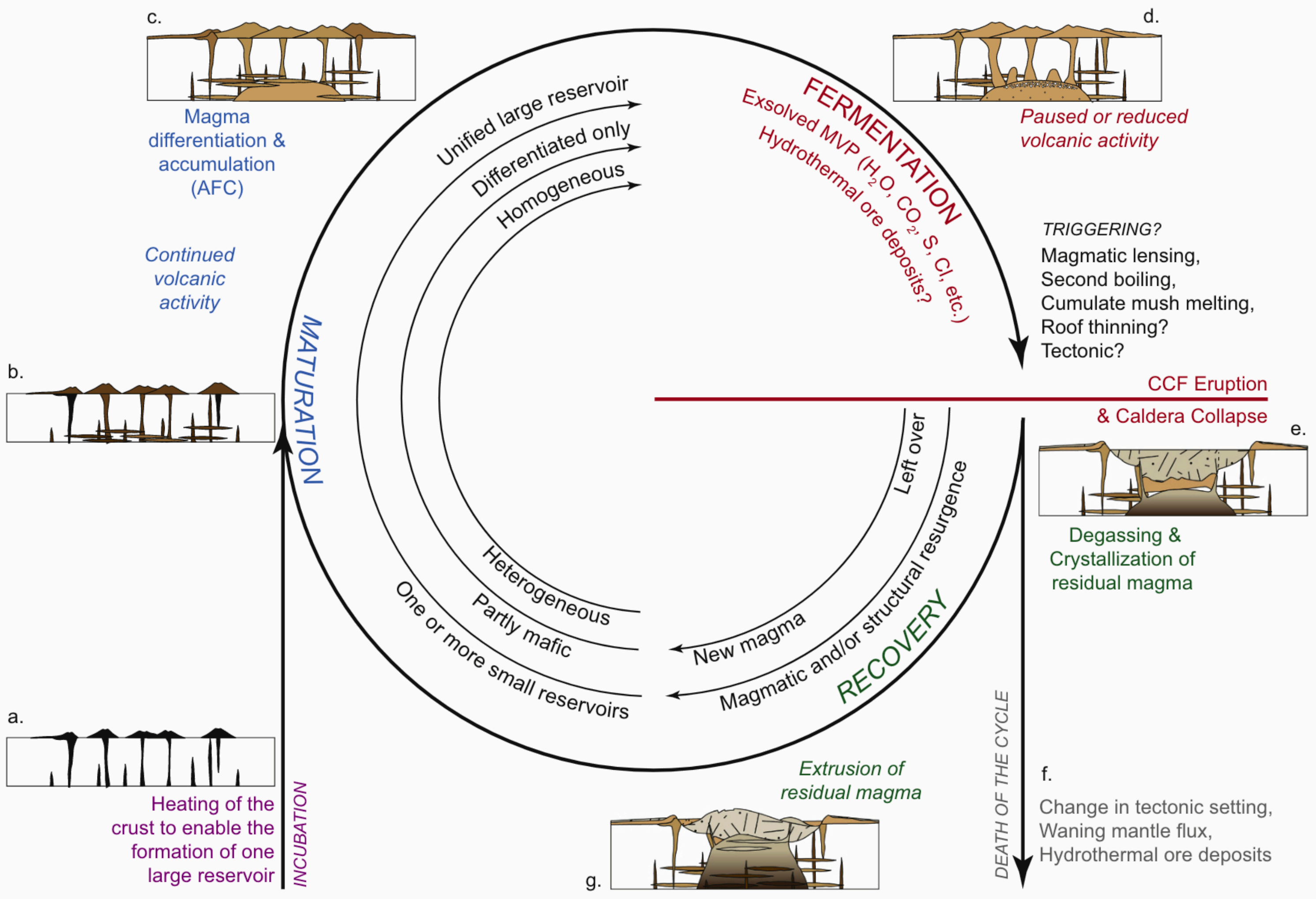
- **Last week**: We learned how to handle data using `Pandas`
  - → Load, access, query, filter, sort and operate on data
- **This week:** You received / compiled / produced a new dataset
- Review common tasks / method to get accustomed to the dataset
  - → Understand its **structure** → what type of data does it contain
  - → Explore it **visually** → plotting with dedicated libraries
  - → Describe **single variables** → univariate analyses
  - → Assess coupled behaviours of **two variables** → bivariate analyses
  - → First attempt at **modelling** these behaviours → linear regressions

The dataset

Case study

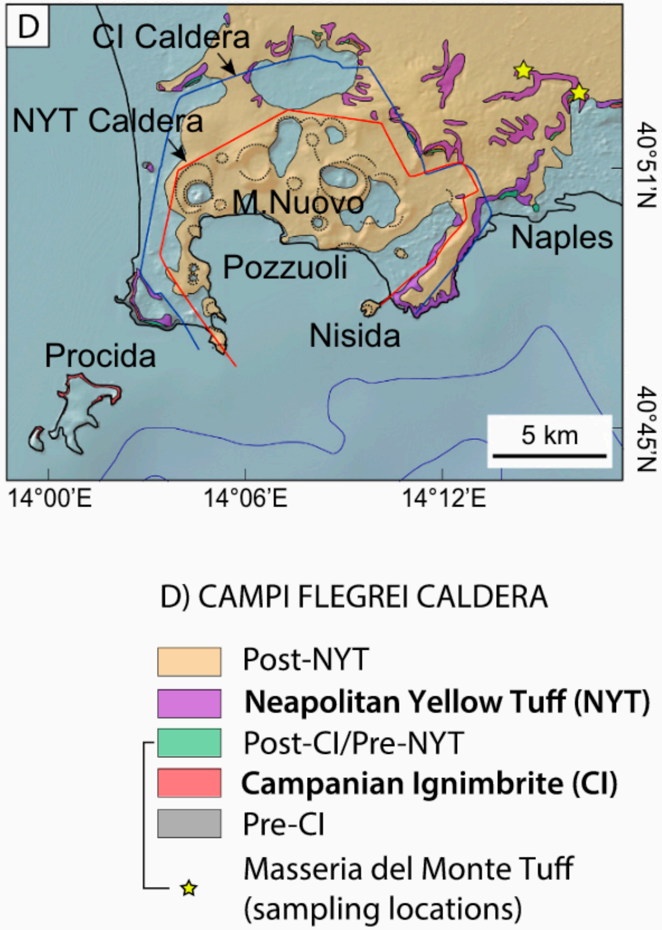
- **Campi Flegrei**: Home to 1.5 million people
- **Caldera-forming** eruptions
  - → Campanian Ignimbrite (CI), ~39 ka ago
  - → Neapolitan Yellow Tuff (NYT), ~15 ka ago

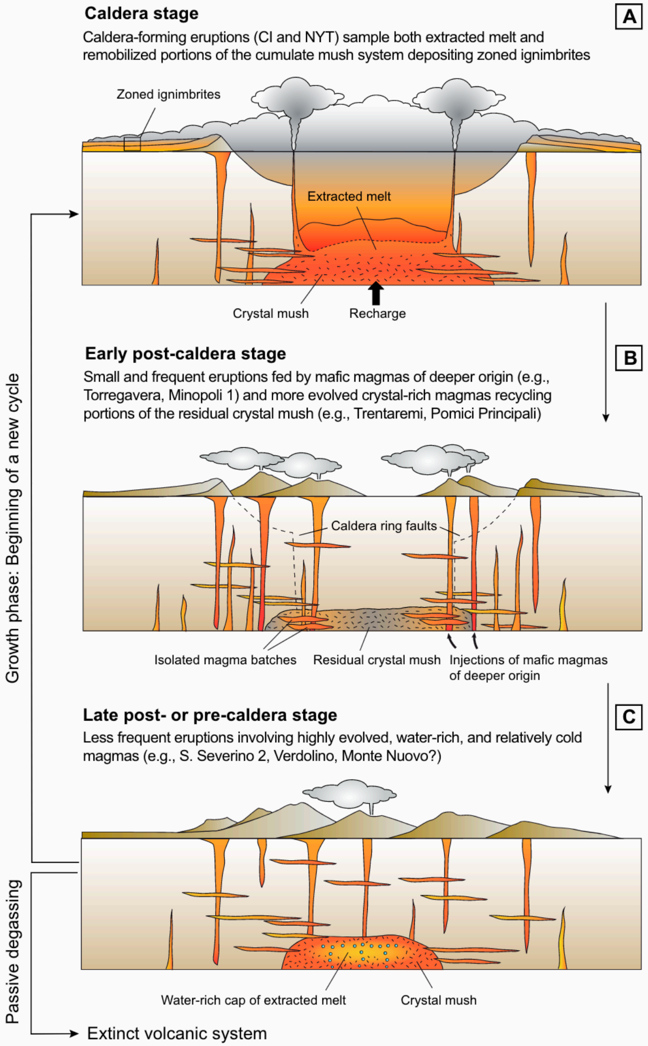
Caldera-forming eruptions cycles



Bouvet de Maisoneuve et al. (2021)

Caldera cycles at Campi Flegrei



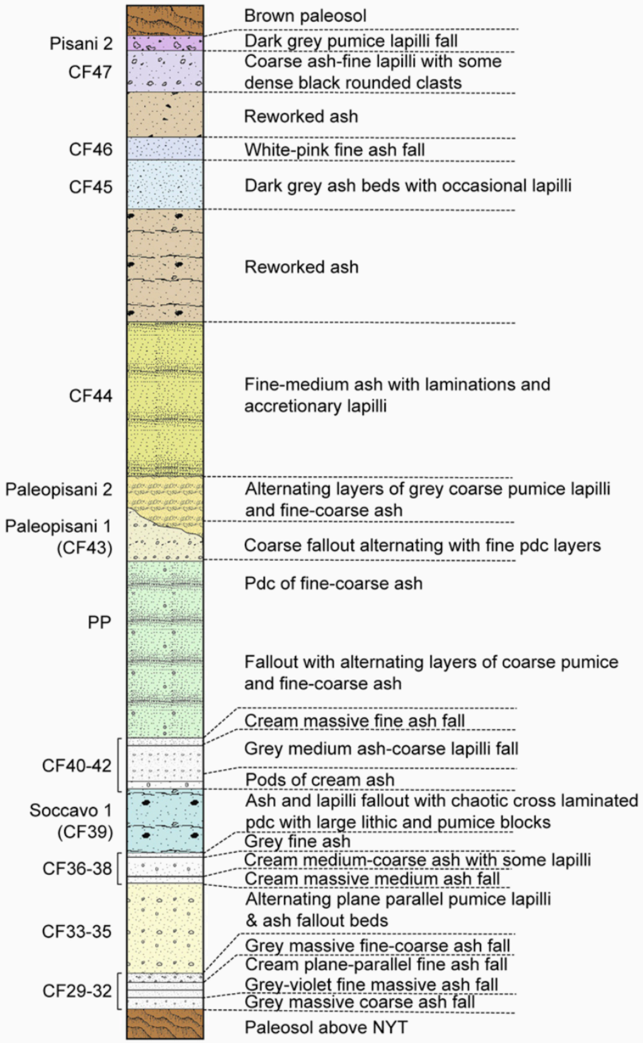


Bouvet de Maisoneuve et al. (2021), Forni et al. (2018) →

Dataset

Tephrostratigraphy and glass compositions of post-15 kyr Campi Flegrei eruptions

- Smith et al. (2011)
- Major and trace elements



Loading the dataset

- Import `pandas`
- Load an **excel** file using `pandas`
- Specify **which sheet** to read from using the `sheet_name` argument
  - → *Supp_majors* → `df_majors`
  - → *Supp_traces* → `df_traces`

```
# Load Pandas
import pandas as pd # Import pandas

# Import the dataset specifying which Excel sheet name to load the data from
df_majors = pd.read_csv('https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data
df_traces = pd.read_csv('https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data
```

Inspecting the dataset

```
df_traces.head(5)
```

	Analysis no.	Strat. Pos.	Eruption	controlcode	Sample	Ep
0	1915	63	Astroni 7	1	79	th b
1	1916	63	Astroni 7	1	79	th b
2	1917	63	Astroni 7	1	79	th b
3	1918	63	Astroni 7	1	79	th b
4	1919	63	Astroni 7	1	79	th b

5 rows × 37 columns

```
df_traces.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 370 entries, 0 to 369
Data columns (total 37 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Analysis no.       370 non-null    int64
 1   Strat. Pos.        370 non-null    int64
 2   Eruption           370 non-null    object
 3   controlcode        370 non-null    int64
 4   Sample             370 non-null    object
 5   Epoch              370 non-null    object
 6   Crater size        370 non-null    int64
 7   Date of analysis   370 non-null    object
 8   Si/bulk cps        370 non-null
float64
 9   SiO2* (EMP)        370 non-null
float64
 10  Sc                 370 non-null
```

Types of data

Basic types of data

- `int`: **Integer** numbers → 0, 1, 2, ...
- `float`: **Decimal** numbers → 1.1, 1.2, 1.3, ...
- `object`: **Strings** → “Campi Flegrei”

A note on *Bytes*

- `int` and `float` numbers are followed by 8, 16, 32 or 64 → **Byte** → control how much memory a variable uses
- A **Bit** is a binary digit → 0/1 → smallest possible unit of data
- `int8` can store 2^8 digits = 256
- `int16` can store $2^1 6$ digits = 65’563

Families of data

Numerical data

- Represent quantities or measurable values
- **Quantitative**
  - → *Discrete*: Earthquake counts
  - → *Continuous*: Temperature
- Stored as **numerical** values
- **Operations** → arithmetic

Categorical data

- Represent categories or groups of data
- **Qualitative** / semi-quantitative
  - → *Nominal*: No order → Landslide type
  - → *Ordinal*: Order → rank
- Stored as **strings** or **integer**
- **Operations** → counting, grouping

Families of data - cheat sheet

Feature	Categorical Data	Numerical Data
Definition	**Categories or groups** of data	**Quantities or measurable values**
Nature	Qualitative (describes qualities)	Quantitative (describes amounts or measurements)
Data Type	Non-numeric (often text or labels)	Numeric (numbers only)
Examples	Eruption type	Element concentration
Possible Operations	Counting, grouping, mode	Arithmetic operations
Measurement Scale	Nominal or Ordinal	Interval or Ratio
Visualization Tools	Bar chart, Pie chart	Histogram, Box plot, Scatter plot
Subtypes	**Nominal:** Categories with no order (e.g., color); **Ordinal:** Categories with order (e.g., rank)	**Discrete:** Countable numbers (e.g., number of earthquakes); **Continuous:** Measurable values (e.g., temperature)
Examples of Statistical Summary	Frequency, mode	Mean, median, standard deviation
Storage Format	Strings or labels	Integers or floats

Plotting data

Plotting libraries

There are two main plotting libraries:

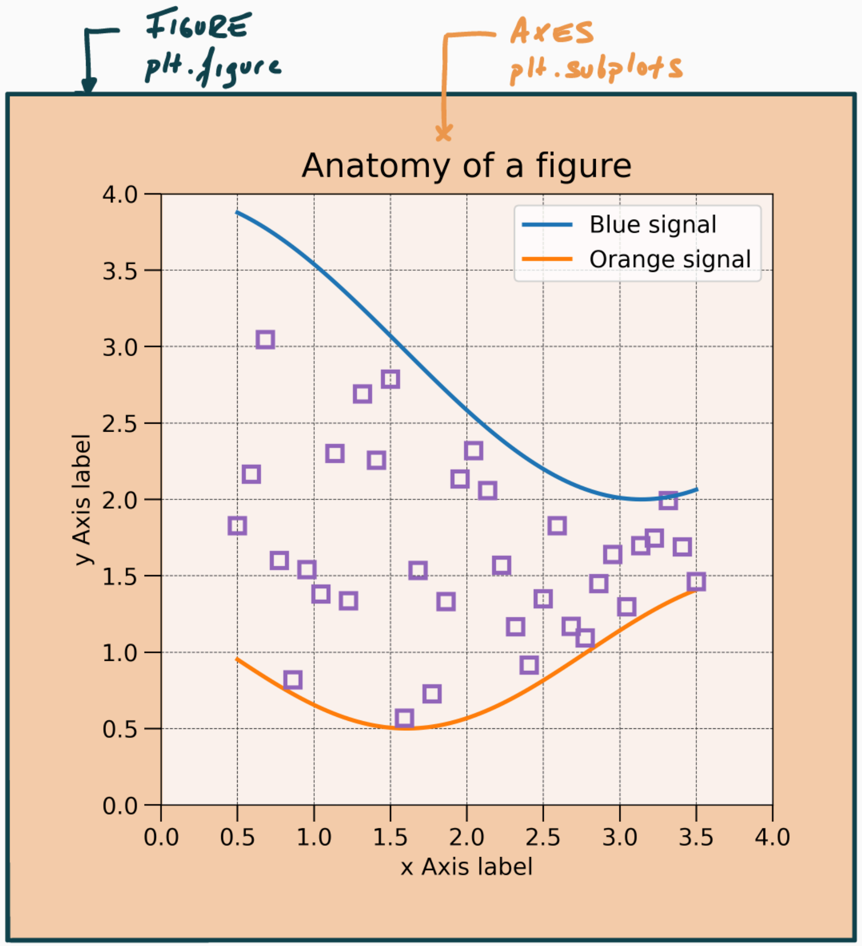
- `matplotlib` (and its module `pyplot`): core of plotting in Python
  - → Matplotlib gallery
- `seaborn`: based on `matplotlib`, designed for statistical exploration, integration with `pandas`
  - → Seaborn gallery

```python
import matplotlib.pyplot as plt # Import the pyplot module from matplotlib
import seaborn as sns # Import seaborn
```

Anatomy of a figure

- Setup figure: `plt.subplots()`
- Returns:
  - → `fig` → figure
  - → `ax` → axes

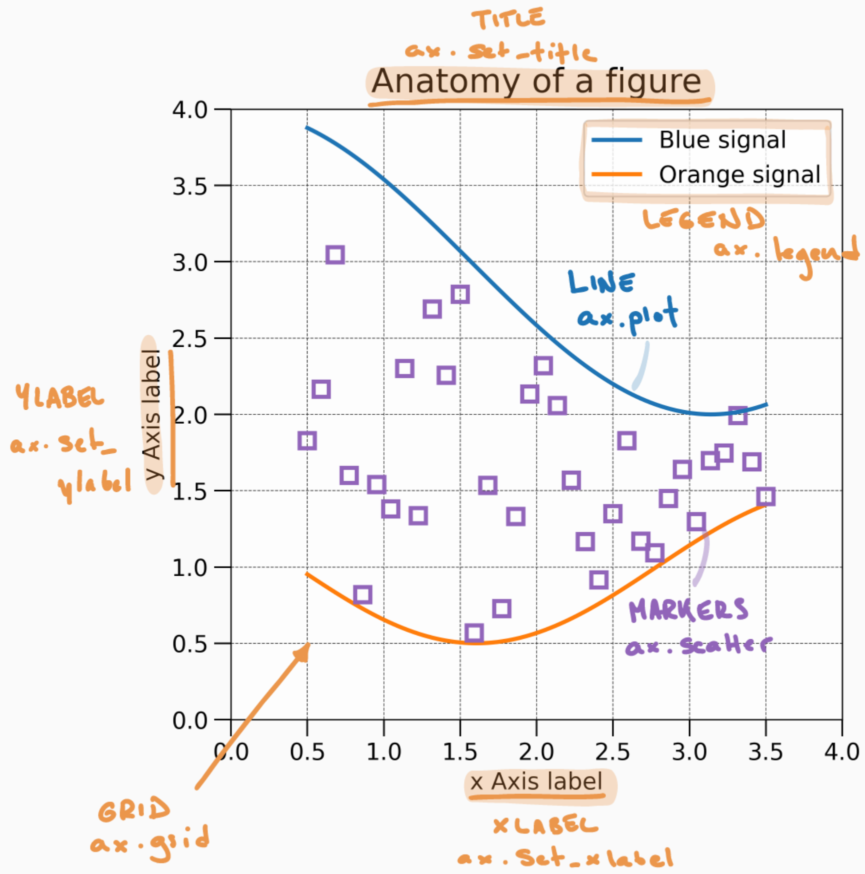
```
fig, ax = plt.subplots()
```



Anatomy of a axes

Axes: Where most of the magic occurs

Function	Description
`ax.set_title`	Sets the title of the axes
`ax.set_xlabel`	Sets the label for the x-axis
`ax.set_ylabel`	Sets the label for the y-axis
`ax.legend`	Displays the legend
`ax.grid`	Shows grid lines

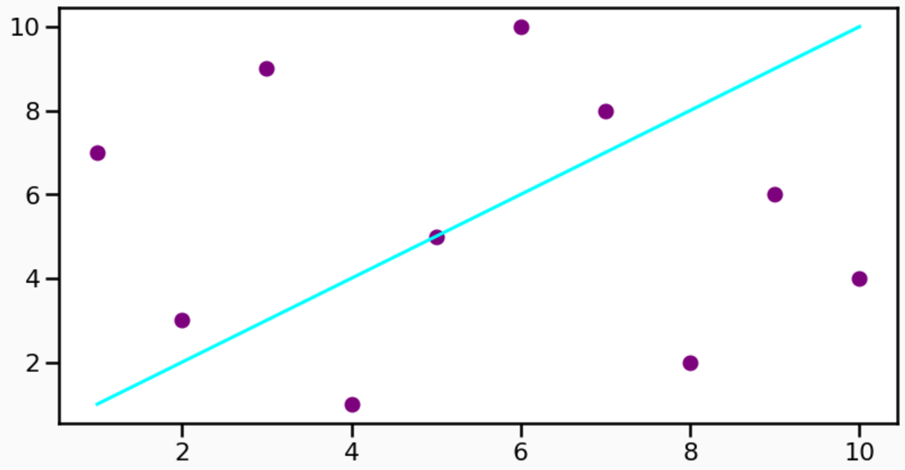


Plotting example

```python
# Define some data
data1 = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
data2 = [7, 3, 9, 1, 5, 10, 8, 2, 6, 4]

# Set the figure and the axes
fig, ax = plt.subplots()

# Plot the data
ax.plot(data1, data1, color='aqua', label='L
ax.scatter(data1, data2, color='purple', lab
```



Plotting example

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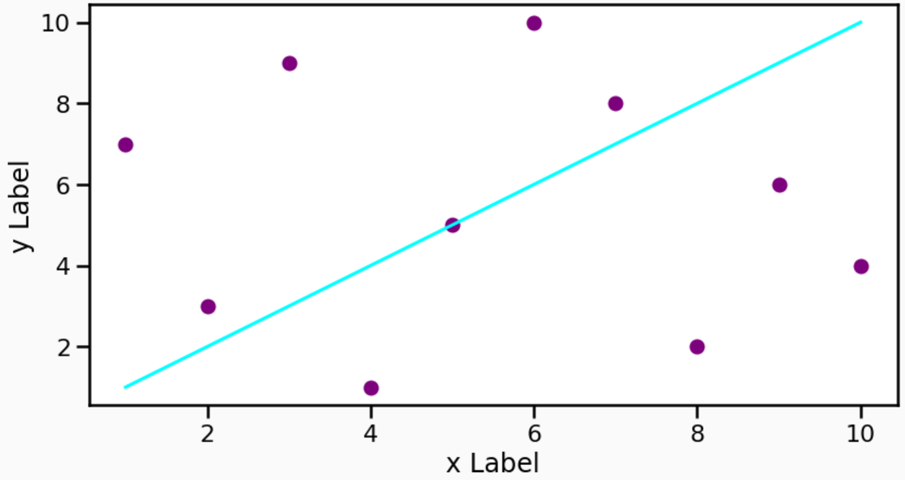
# Define some data
data1 = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
data2 = [7, 3, 9, 1, 5, 10, 8, 2, 6, 4]

# Set the figure and the axes
fig, ax = plt.subplots()

# Plot the data
ax.plot(data1, data1, color='aqua', label='L
ax.scatter(data1, data2, color='purple', lab

# Set labels
ax.set_xlabel('x Label')
ax.set_ylabel('y Label')

# Only to make it look pretty on the present
plt.show()
```



Seaborn

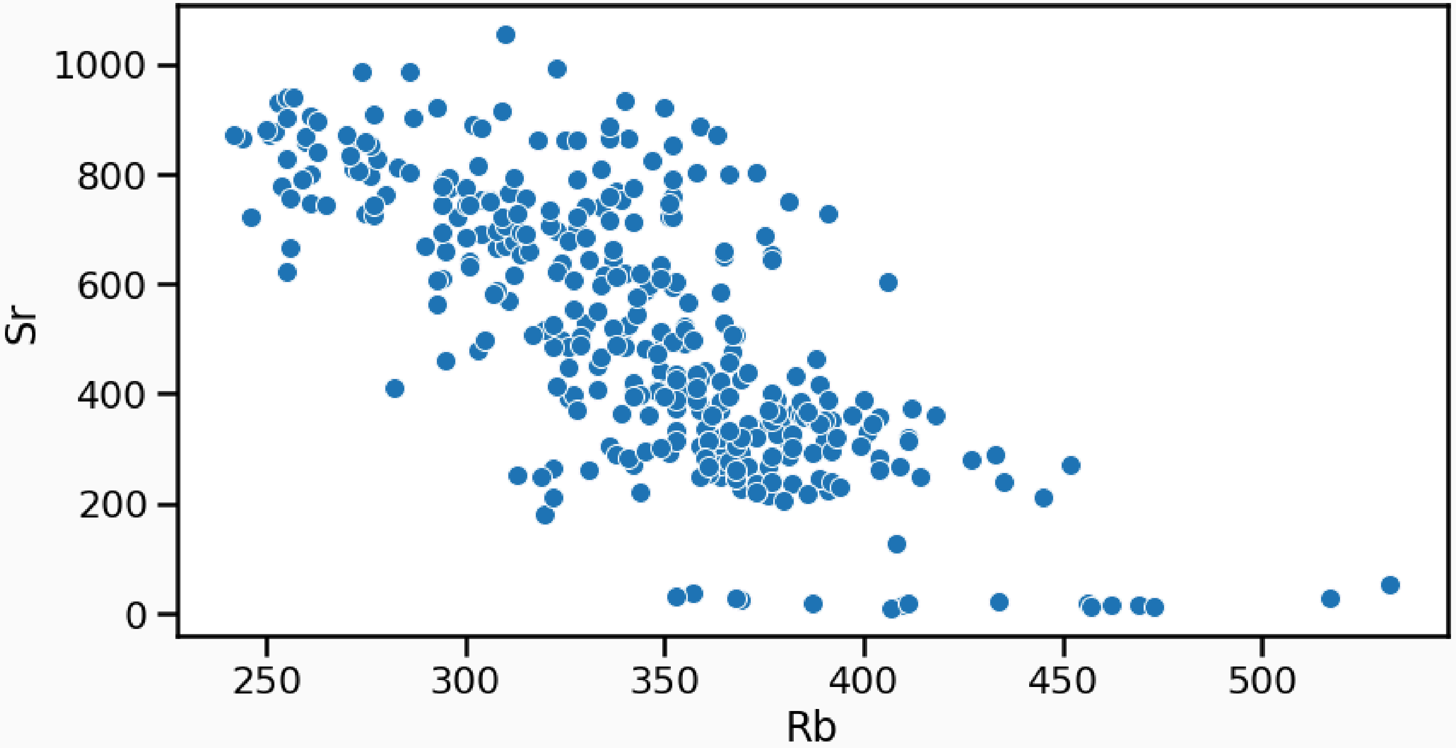
Why Seaborn?

> Seaborn provides a high-level interface for drawing attractive and informative statistical graphics.

- Makes plotting `pandas` DataFrames **easy**
- Produces **clean** graphics
- Drawback: difficult customisation

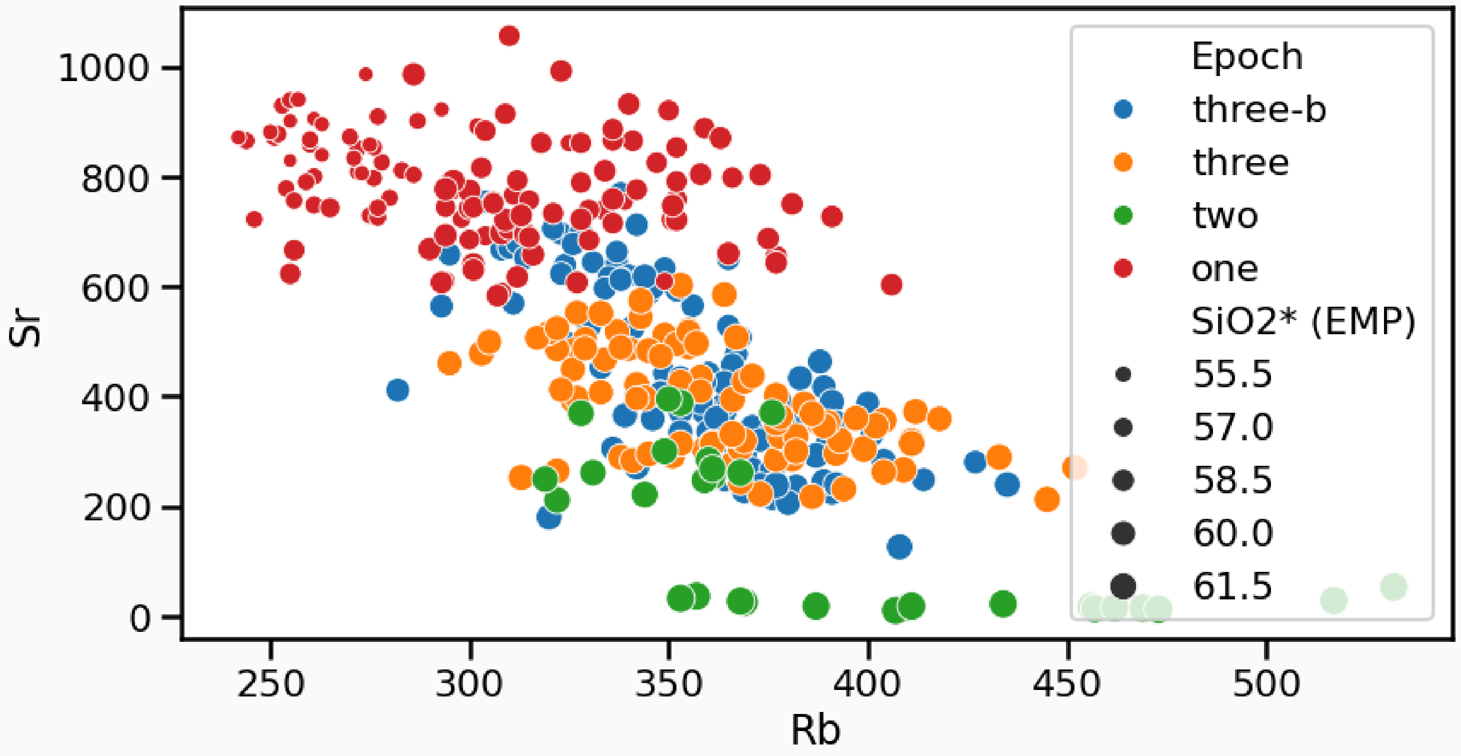
Seaborn example

```python
# Define figure + axes
fig, ax = plt.subplots()
# Plot with seaborn (remember, we imported it as sns)
# df_traces is the geochemical dataset imported previously
sns.scatterplot(ax=ax, data=df_traces, x='Rb', y='Sr')
```



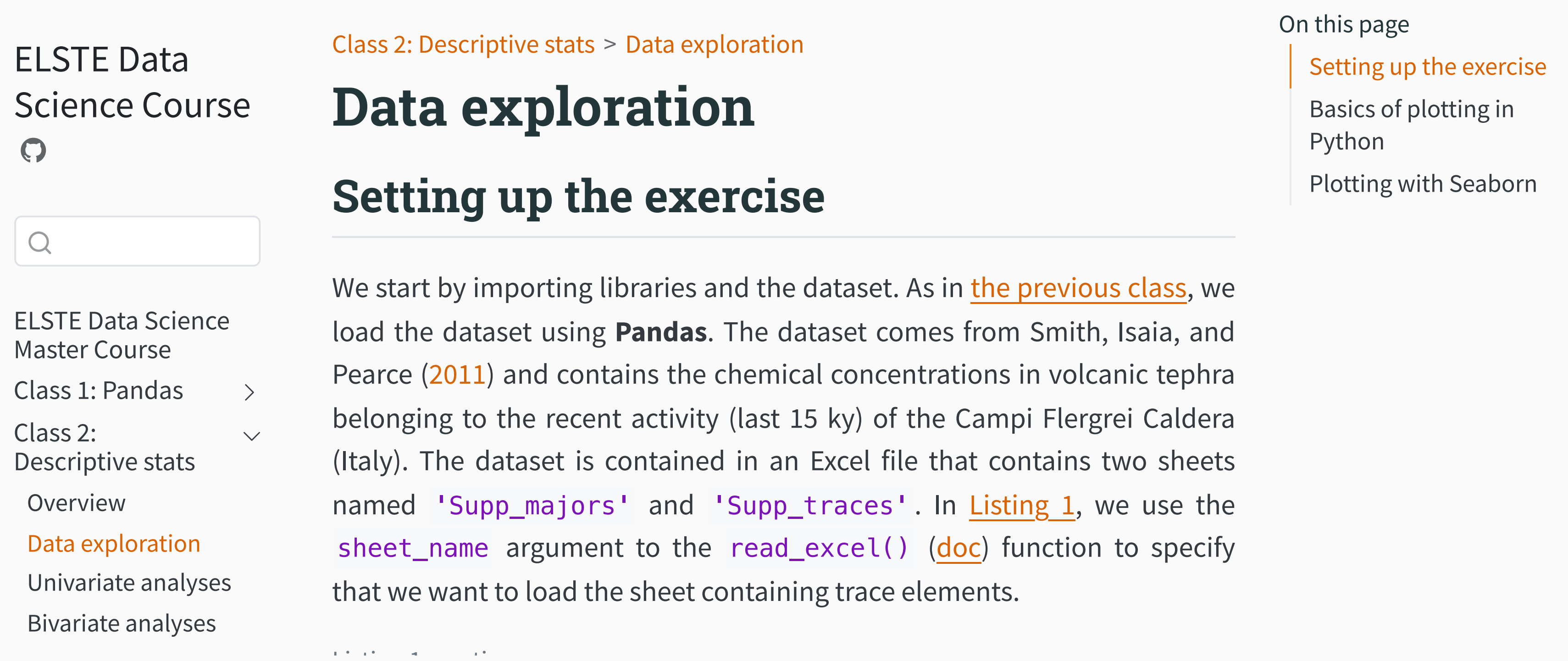
Seaborn example

```python
# Define figure + axes
fig, ax = plt.subplots()
# Plot with seaborn (remember, we imported it as sns)
# df_traces is the geochemical dataset imported previously
sns.scatterplot(ax=ax, data=df_traces, x='Rb', y='Sr', hue='Epoch', size="SiO2* (EMP)")
```



Your turn!

- Go to https://elste-master.github.io/Data-Science/
  - → Class 2 > Data exploration
  - → Explore the structure of the glass geochemistry or your dataset
  - → Get used to plotting functions


ELSTE Data Science Course

ELSTE Data Science Master Course
Class 1: Pandas
Class 2: Descriptive stats
Overview
Data exploration
Univariate analyses
Bivariate analyses

Class 2: Descriptive stats > Data exploration

Data exploration

Setting up the exercise

We start by importing libraries and the dataset. As in the previous class, we load the dataset using **Pandas**. The dataset comes from Smith, Isaia, and Pearce (2011) and contains the chemical concentrations in volcanic tephra belonging to the recent activity (last 15 ky) of the Campi Flergrei Caldera (Italy). The dataset is contained in an Excel file that contains two sheets named `'Supp_majors'` and `'Supp_traces'`. In Listing 1, we use the `sheet_name` argument to the `read_excel()` (doc) function to specify that we want to load the sheet containing trace elements.

On this page
Setting up the exercise
Basics of plotting in Python
Plotting with Seaborn

Descriptive statistics

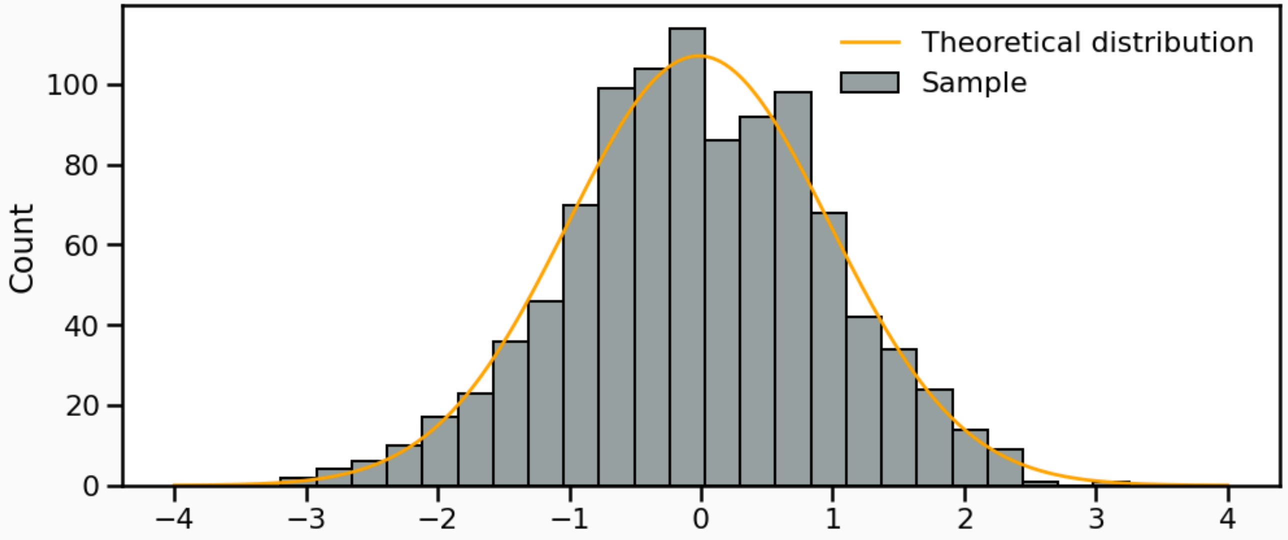
Descriptive statistics

- **Descriptive statistics** → provide ways of capturing the properties of a given data set or sample.
- `Pandas` and `Seaborn` → review metrics, tools, and strategies to summarize a dataset
- Providing us with a quantitative basis to describe it and compare it to other datasets

Univariate analyses

Univariate analyses

- **Objective** capturing the properties of **single** variables at the time
- Fist step: review the samples distribution using **histograms**
  - → Divides dataset into **equal intervals** → bins
  - → **Counts the value** in each bin
  - → Represents the distribution of data

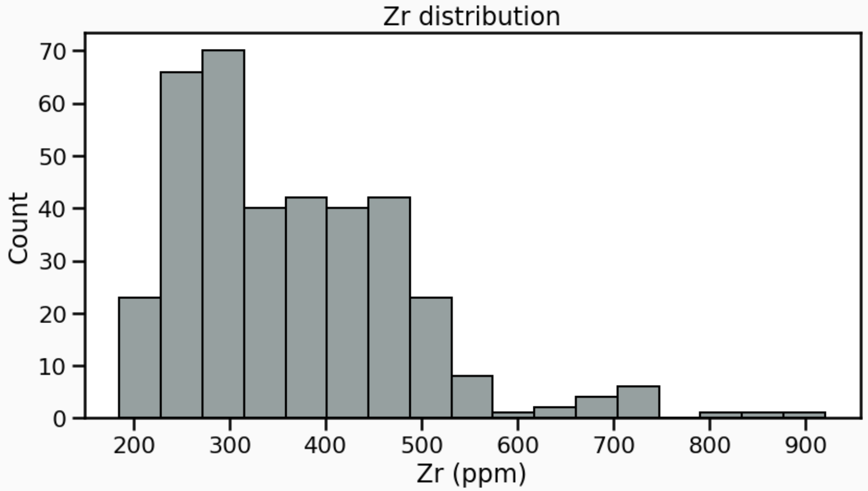


Univariate analyses

- **Objective** capturing the properties of **single** variables at the time
- Fist step: review the samples distribution using **histograms**
  - → Divides dataset into **equal intervals** → bins
  - → **Counts the value** in each bin
  - → Represents the distribution of data

Automatic number of bins:

```
fig, ax = plt.subplots()
# Plot histogram
sns.histplot(data=df_traces, x='Zr', color
ax.set_xlabel('Zr (ppm)');
ax.set_title('Zr distribution');
```

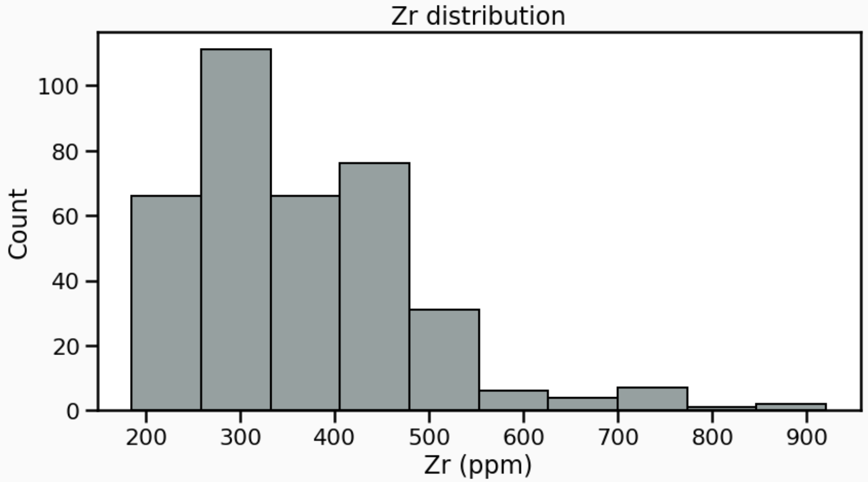


Univariate analyses

- **Objective** capturing the properties of **single** variables at the time
- Fist step: review the samples distribution using **histograms**
  - → Divides dataset into **equal intervals** → bins
  - → **Counts the value** in each bin
  - → Represents the distribution of data

10 bins:

```
fig, ax = plt.subplots()
# Plot histogram
sns.histplot(data=df_traces, x='Zr', bins=
ax.set_xlabel('Zr (ppm)');
ax.set_title('Zr distribution');
```

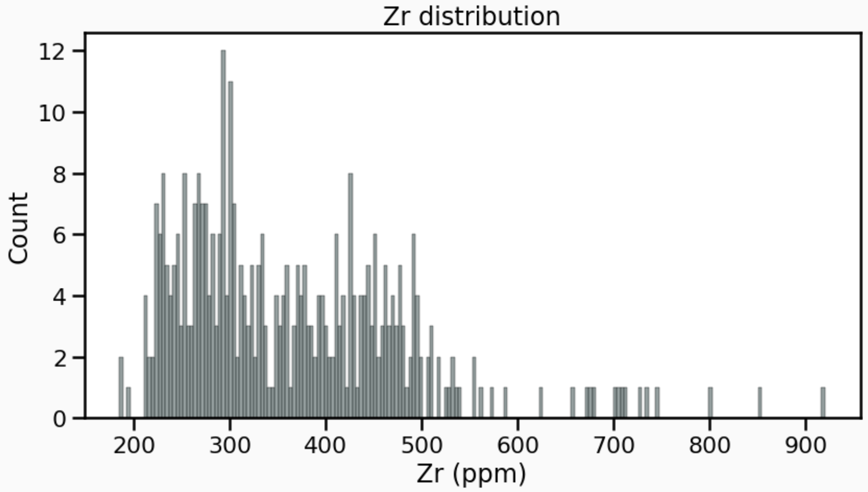


Univariate analyses

- **Objective** capturing the properties of **single** variables at the time
- Fist step: review the samples distribution using **histograms**
  - → Divides dataset into **equal intervals** → bins
  - → **Counts the value** in each bin
  - → Represents the distribution of data

200 bins:

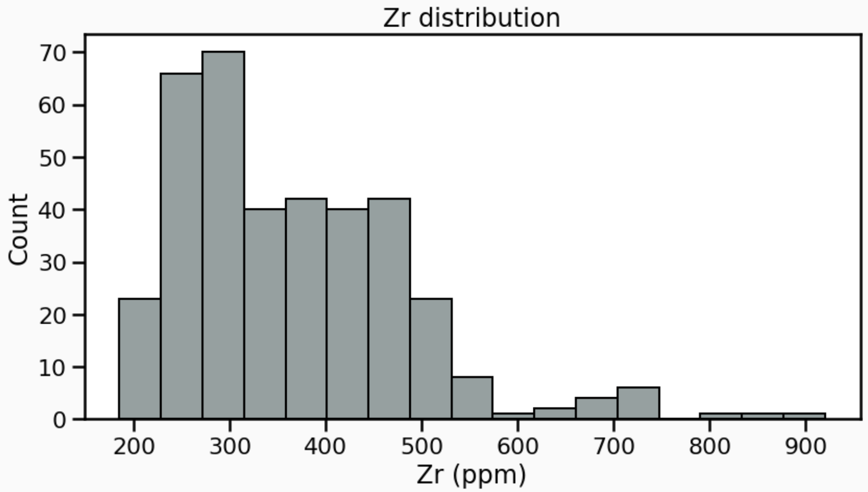
```
fig, ax = plt.subplots()
# Plot histogram
sns.histplot(data=df_traces, x='Zr', bins=
ax.set_xlabel('Zr (ppm)');
ax.set_title('Zr distribution');
```



Descriptive parameters

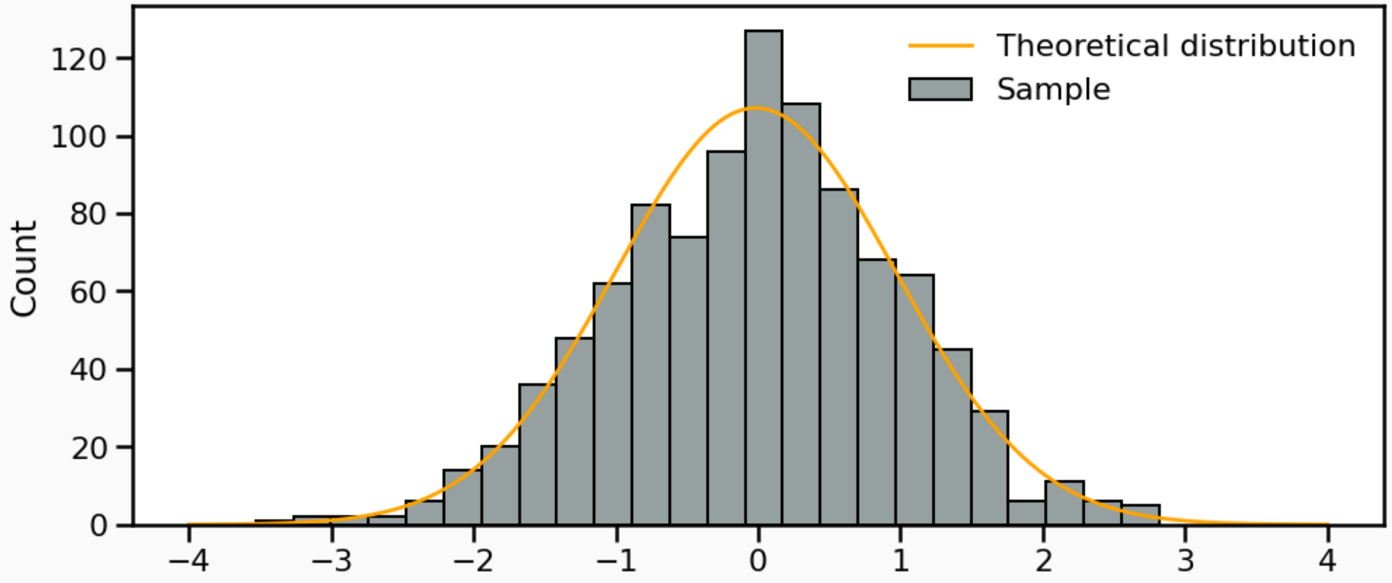
Intuitive importance of describing **three different characteristics** of the distribution:

- The **location** - or where is the central value(s) of the dataset;
- The **dispersion** - or how spread out is the distribution of data compared to the central values;
- The **skewness** - or how symmetrical is the distribution of data compared to the central values;



Descriptive parameters

- **Ideally** → able to constrain full distributions of X
  - → **theoretical moments**
- **In practice** → only observe a finite sample of X
  - → **estimators**



Location I: Mean

The **mean** is the sum of the values divided by the number of values:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Meaningful to describe symmetrical distributions
- Very **sensitive** to outliers
- `df.mean()`

```python
# One column
df_traces[['Zr']].mean().round(2)
# Multiple columns
df_traces[['Zr','Rb','Sr']].mean().round(2)
```

```
Zr     365.39
Rb     343.81
Sr     516.42
dtype: float64
```

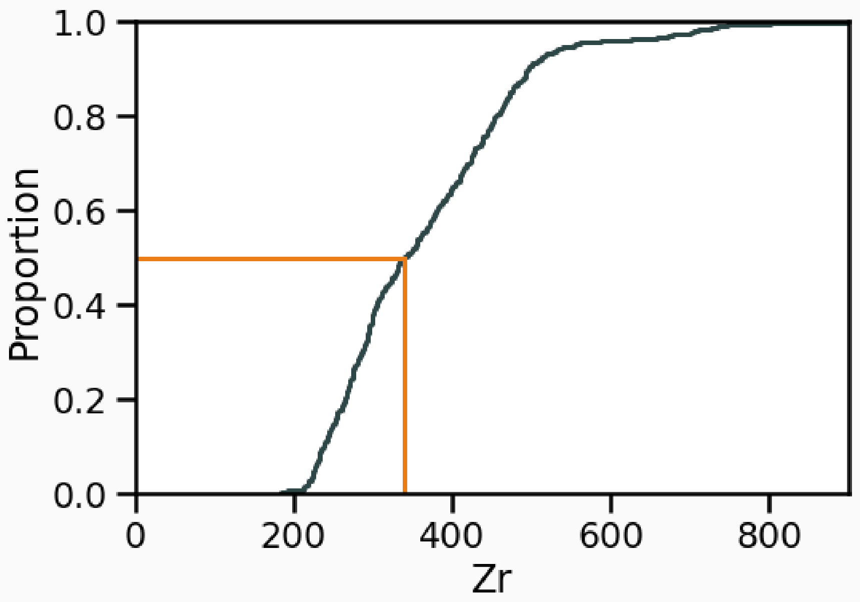
Location II: Median

The **median** is the value at the exact middle of the dataset.

- No equation, use of ECDF
- `df.median()`

```python
# One column
df_traces[['Zr']].median().round(2)
# Multiple columns
df_traces[['Zr','Rb','Sr']].median().round
```

```
Zr    339.0
Rb    347.5
Sr    490.0
dtype: float64
```

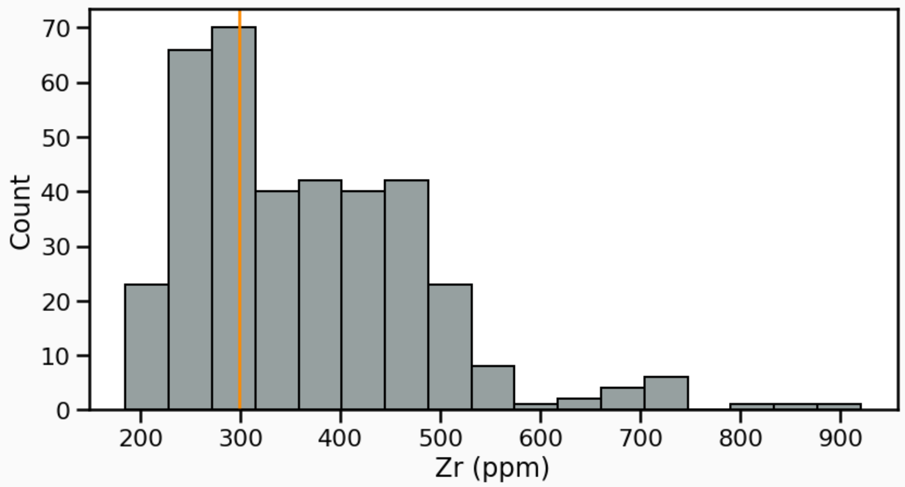


Location III: Mode

The **mode** is the value that occur most frequently in a dataset.

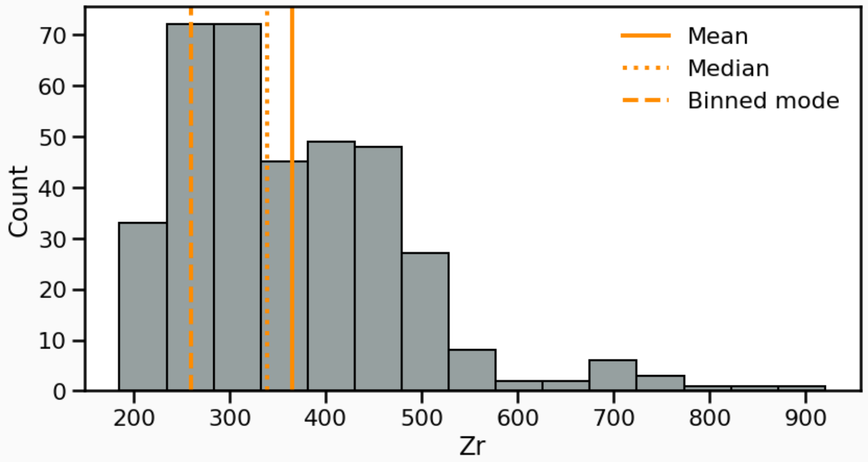
- **Categorical** or **discrete numerical data**: value(s) that appear most often.
- For **continuous data**, exact duplicates may be rare

```python
# Calculate the modes
modes = df_traces['Zr'].mode().round(2)

# Plot the histogram
fig, ax = plt.subplots()
sns.histplot(data=df_traces, x='Zr', ax=ax,

# Plot the modes
[ax.axvline(m, color='darkorange', lw=2) for

ax.set_xlabel('Zr (ppm)')
plt.show()
```



Location: Summary

- **Mean**: Best for symmetric distributions
  - → Sensitive to outliers
- **Median**: Best for skewed or outlier-prone data
  - → Only based on rank, not magnitude
- **Mode**: Best for describing the most frequent range in grouped or categorical-like data.
  - → Otherwise requires binning



Dispersion 1: Range

The **range** is the range of value covered in the dataset

- Min → `df.min()`
- Max → `df.max()`
- Range:

$$\text{Range} = \max(x) - \min(x)$$

```python
df_min = df_traces['Zr'].min()
df_max = df_traces['Zr'].max()
df_range = df_max - df_min
print(f"The range of Zircon concentrations
```

```
The range of Zircon concentrations is 735.00
```

> ⚠ **Don't use Python reserved keywords as variable names**
>
> Names as `min`, `max` or `range` represent native Python variables and **cannot be used as variable names!**

Dispersion 2: Standard deviation

The **sample standard deviation** (s) is the sum of squares differences between data points x_i and the mean $\bar{x}$ over n observations:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

- s is in the same unit as the data
  - → Easy to interpret
- `df.std()`

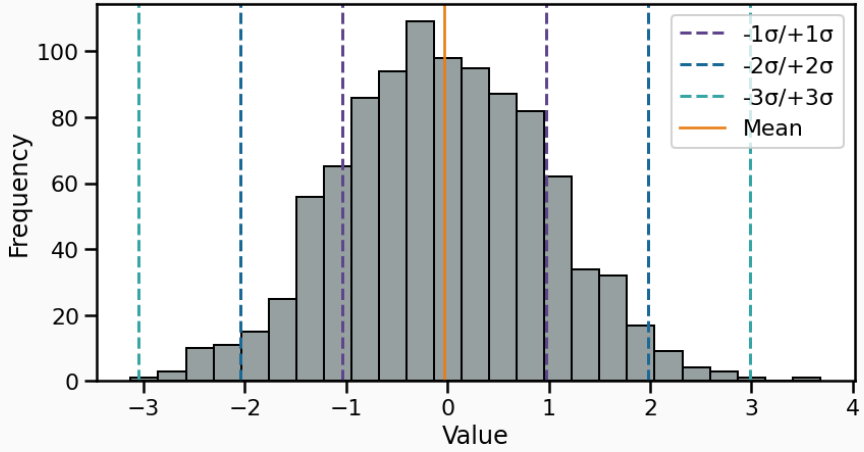
```python
df_traces['Zr'].std()
```

```
118.39358315895393
```

Dispersion 2: Standard deviation

The **theoretical standard deviation** (σ) is closely related to the **normal distribution** as:

- 1σ → ~68% of the data
- 2σ → ~95% of the data
- 3σ → ~99.7% of the data



Dispersion 2.1: Variance

The **variance** (s^2) is the average of squared deviations from the mean (→ the square of the standard deviation):

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

- s^2 is in **not** the same unit as the data
- Used in modelling (e.g., ANOVA)
- `df.var()`

```python
df_traces['Zr'].var()
```

```
14017.04053321614
```

Rule of thumb

- Comparing data spread → standard deviation
- Statistical modelling → variance

Dispersion 3: Interquartile range

The **interquartile range** (IQR) indicates how spread out the middle 50% of the data is.

- Difference between the third quartile (Q3) and the first quartile (Q1):

$$\mathrm{IQR} = Q_3 - Q_1 \qquad (1)$$

- Q1: The value below which **25% of the data fall**
- Q3: The value below which **75% of the data fall**

Dispersion 3: Interquartile range

Table 1: Relationship between quartiles, deciles and percentiles.

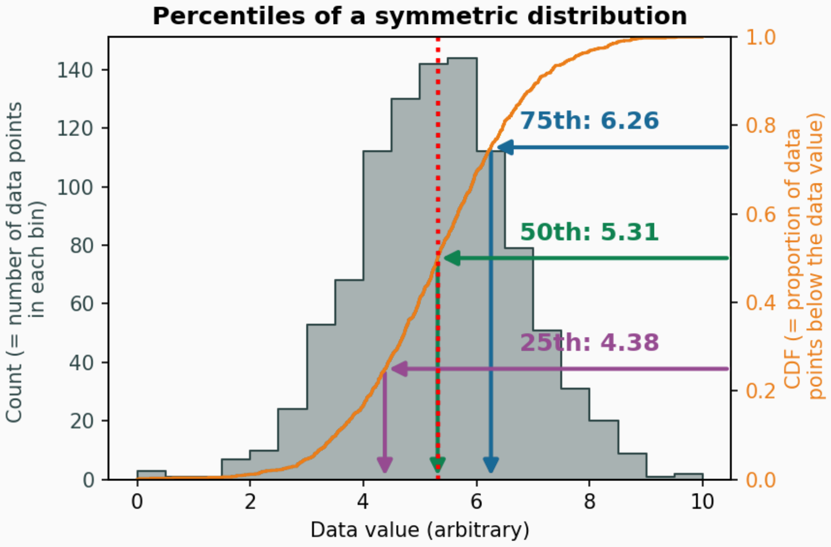
Measure	Number of Divisions	Position(s) in Data / Percentile Equivalent
Quartiles (Q1, Q2, Q3)	4 equal parts	Q1 = 25th percentile, Q2 = 50th percentile (median), Q3 = 75th percentile
Deciles (D1 … D9)	10 equal parts	D1 = 10th percentile, D2 = 20th percentile, …, D9 = 90th percentile
Percentiles (P1 … P99)	100 equal parts	P1 = 1st percentile, P2 = 2nd percentile, …, P99 = 99th percentile

- median =
  - → 2nd quartile = Q2
  - → 5th decile = D5
  - → 50th percentile = P50

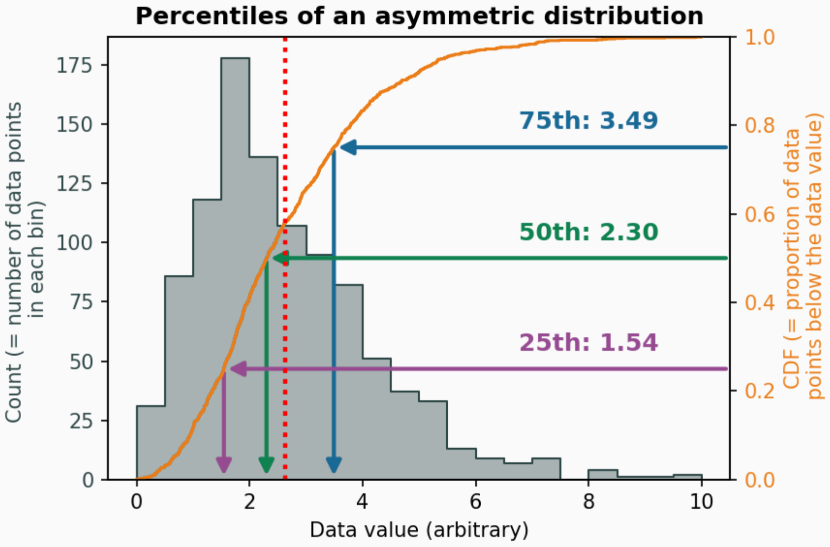
Dispersion 3: Interquartile range

The **interquartile range** (IQR) indicates how spread out the middle 50% of the data is.

Q_1 and Q_3 for a **symmetrical** distribution:



Q_1 and Q_3 for an **asymmetrical** distribution:



Shape: Skewness

The **skewness** measures the asymmetry of a distribution and is computed as:

$$\text{Skewness} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

In *Pandas*, the skewness can be computed using `.skew()`

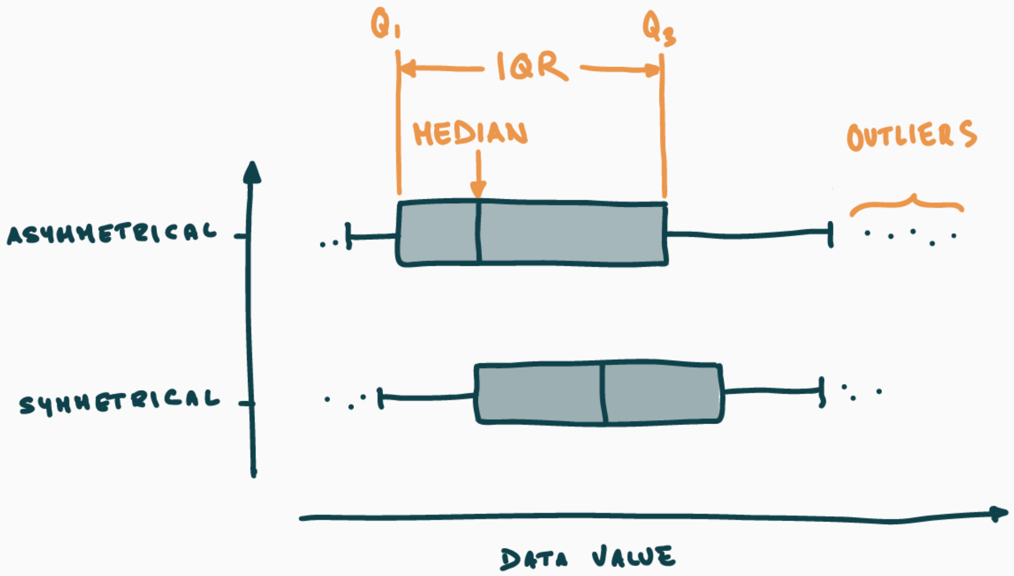
- **0**: Symmetric → e.g., normal distribution
- **>0**: Righ-skewed → a tail extends to the right
- **<0**: Left-skewed → a tail extends to the left

Distribution visualisation

Beyond **histograms** → some plot types report empirical indications of **location**, **dispersion** and **skewness**.

Box and whisker plots

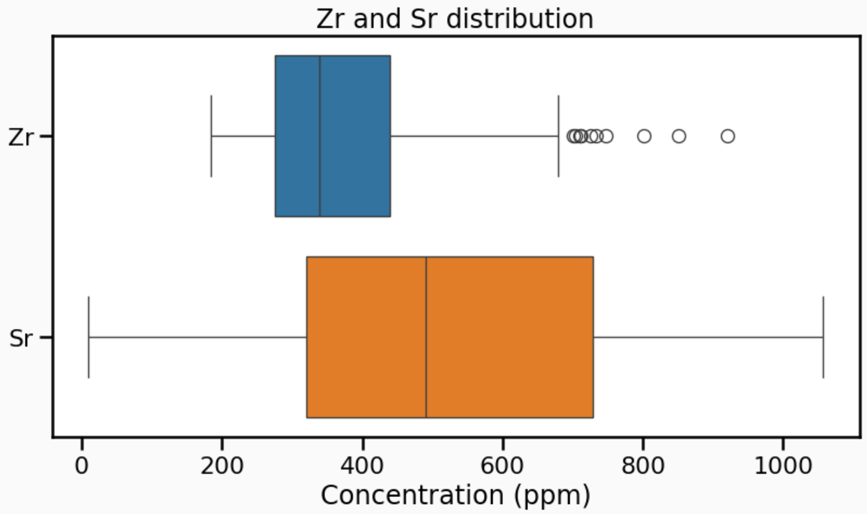
- **Box** → IQR + median
- **Whiskers** 1.5 times the IQR from Q1/Q3
- **Outlier** → differs from the majority of observations



Distribution visualisation: Box plots

- `sns.boxplot()`
- No density
- No tail resolution

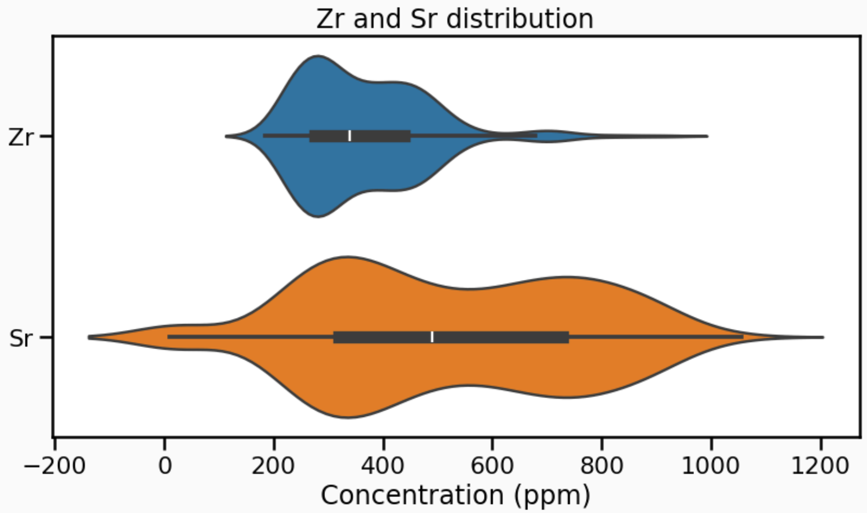
```
fig, ax = plt.subplots()
sns.boxplot(data=df_traces[['Zr', 'Sr']],
ax.set_xlabel('Concentration (ppm)');
ax.set_title('Zr and Sr distribution');
```



Distribution visualisation: Violin plots

- `sns.violinplot()`
- Density estimate through Kernel Density Estimator → artefacts
- Some tail resolution, not always realistic

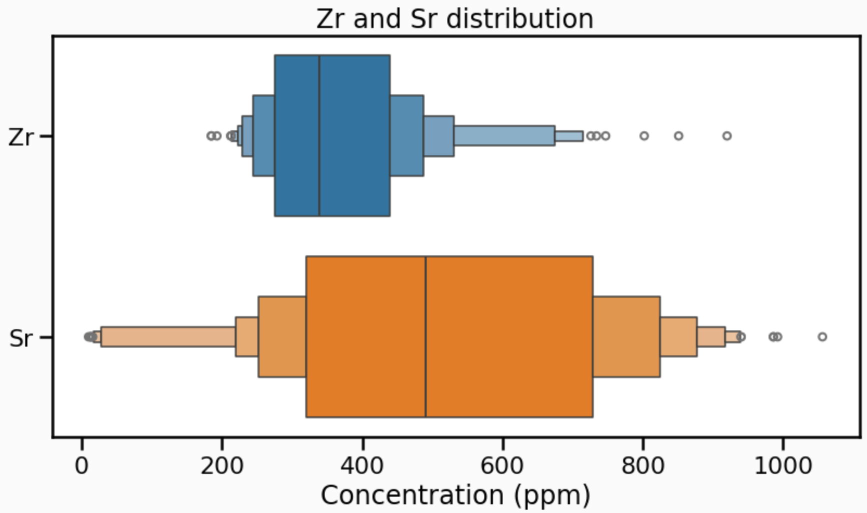
```
fig, ax = plt.subplots()
sns.violinplot(data=df_traces[['Zr', 'Sr']
ax.set_xlabel('Concentration (ppm)');
ax.set_title('Zr and Sr distribution');
```



Distribution visualisation: Boxen plots

- `sns.boxenplot()`
- Plots deciles → some density estimate
- Good tail resolution

```
fig, ax = plt.subplots()
sns.boxenplot(data=df_traces[['Zr', 'Sr']]
ax.set_xlabel('Concentration (ppm)');
ax.set_title('Zr and Sr distribution');
```



Your turn!

- Go to https://elste-master.github.io/Data-Science/
  - → Class 2 > Univariate analyses
  - → Explore the structure of the glass geochemistry or your dataset
  - → Get used to the various functions

ELSTE Data Science Course

ELSTE Data Science Master Course
Class 1: Pandas
Class 2: Descriptive stats
Overview
Data exploration
Univariate analyses
Bivariate analyses

Class 2: Descriptive stats > Univariate analyses

Univariate analyses

The objective of **descriptive statistics** is to provide ways of capturing the properties of a given data set or sample. Using *Pandas* and *Seaborn*, we will review some metrics, tools, and strategies that can be used to summarize a dataset, providing us with a quantitative basis to talk about it and compare it to other datasets.

The first aspect of descriptive statistics is **univariate analyses**, which focus on capturing the properties of **single** variables at the time. We are not yet concerned in characterising the relationships between two or more variables. The following sections introduce basic functions to visually analyse single variables and the most important parameters to describe them

On this page
Plotting data distributions
Descriptive parameters
Non-parametric distribution visualisation
Grouping variables

Bivariate analyses

Bivariate analyses

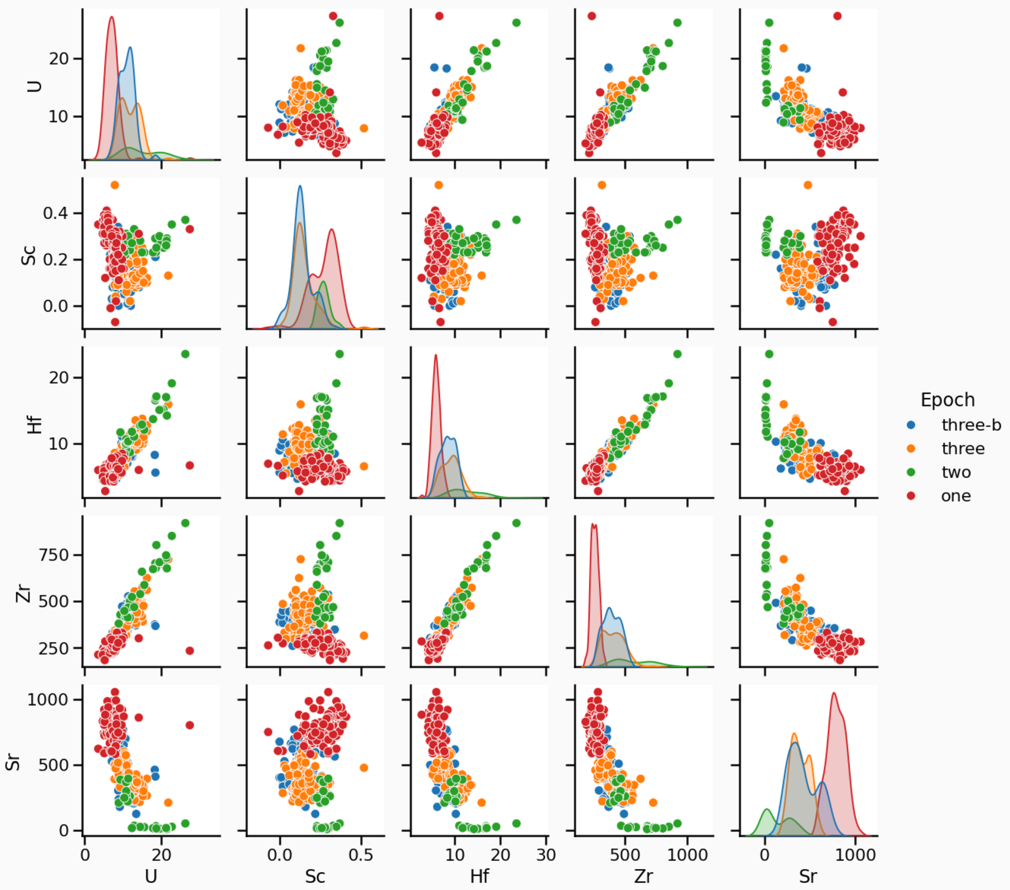
- **Univariate analyses**: properties of one variable at the time
- **Bivariate analyses**: investigate the **relationship** between two variables

1. **Visually** → pairwise comparison
2. Using **covariance/correlation** → correlation matrix
3. First look at **linear regression** → only visually for now

Pairwise comparison

A first visual inspection using `sns.pairplot()`

- Numerical values
- One categorical value



Covariance

- The covariance captures **the direction and magnitude of the linear relationship between the two variables**
- The sample covariance is an **estimator** of the theoretical covariance:

$$\mathrm{Cov}(X, Y)_{\mathrm{emp}} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}),$$

- **Positive**: Increase in $X \rightarrow$ increase in Y
- **Negative**: Increase in $X \rightarrow$ decrease in Y

Problem: depends on the magnitudes of the variables, so it does not directly reflect the strength of the relationship

Correlation

- The correlation coefficient provides **a normalized version** of the covariance
  - → Ranges from -1 to 1
  - → Indicates both the direction and the strength of the linear relationship
- The sample correlation is computed as:

$$r(X,Y)_{\text{emp}} = \frac{\text{Cov}(X,Y)_{\text{emp}}}{s_X s_Y} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}},$$

- $r(X,Y) = -1/1 \rightarrow$ “perfect” **relationship** between X and Y
- $r(X,Y) = 0 \rightarrow$ **independence** between X and Y.

Correlation matrix

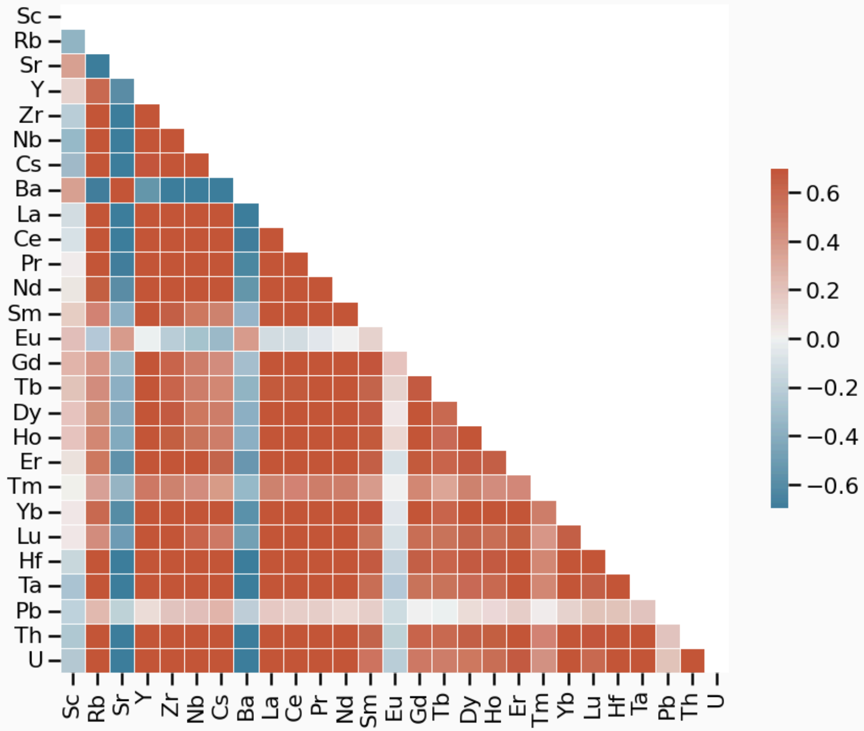
- Correlation matrix → square table showing the pairwise correlations between all variables in a dataset
- Suppose we have p variables $(X_1, X_2, \dots, X_p)$ → the correlation matrix $\mathbf{C}$ is:

$$\mathbf{C} = \begin{pmatrix} 1 & r(X_1, X_2) & \dots & r(X_1, X_p) \\ r(X_2, X_1) & 1 & \dots & r(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ r(X_p, X_1) & r(X_p, X_2) & \dots & 1 \end{pmatrix}.$$

Correlation matrix

Correlation matrix for trace elements

- Use **sample** correlation coefficient r
  - → **Red:** positive relationship
  - → **Blue:** negative relationship
  - → **White:** no relationship



Linear regression

- Correlation coefficients → measure of coupling between two continuous variables
- linear regressions attempt modelling this relationship

$$y = a + bx + \varepsilon$$

where:

- y: dependent (response) variable
- x: independent (predictor) variable
- a: intercept → value of y when $x = 0$
- b: slope → how much y changes by unit of x
- ε: error term

→ **Objective:** Estimate the values of a and b that best fit the observed data

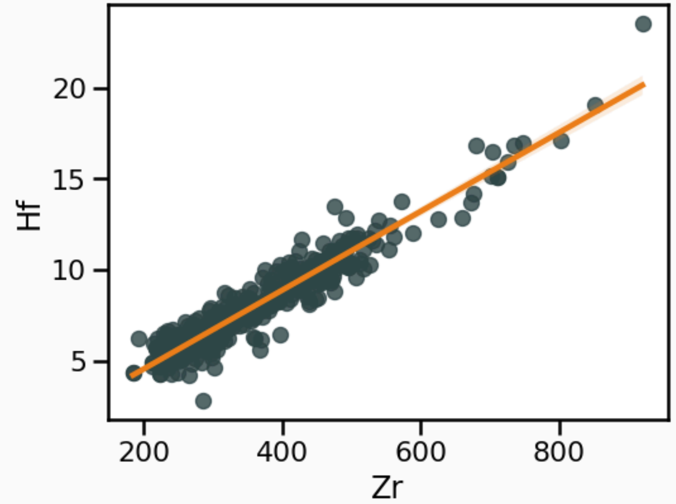
Linear regression with Seaborn

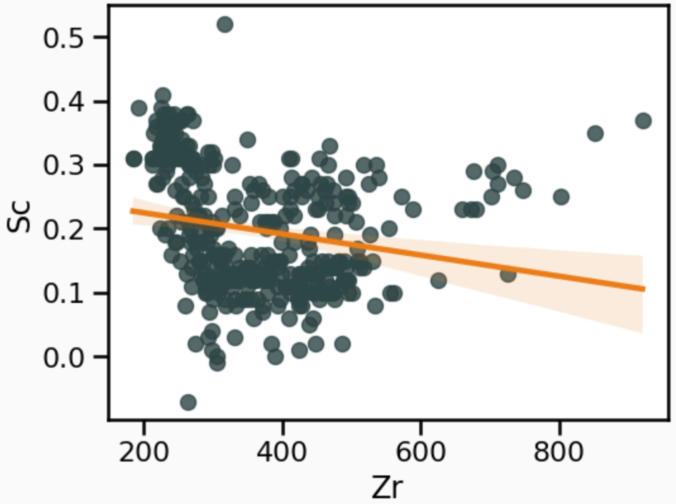
Seaborn has high level functions to **visualise** regressions → `sns.regplot()` that produce:

1. A scatter plot;
2. A linear regression model fit;
3. A confidence interval

→ **Be critical**

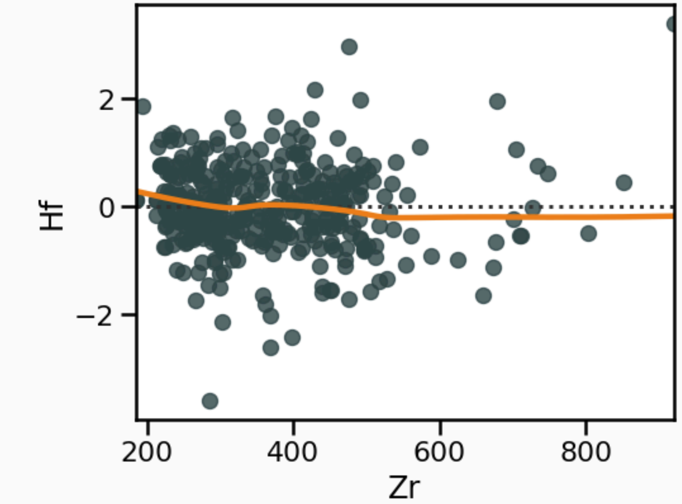
→ Further investigations require dedicated stats packages

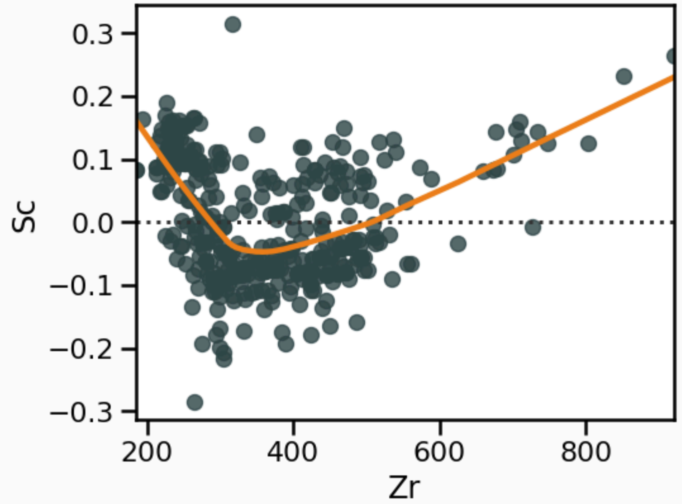




Residual analysis with Seaborn

Linear regression → do residuals show any structure?? → `sns.residplot()`





Residuals are necessary to:

1. **Diagnose model fit** (→ how much **unexplained variation** remains);
2. **Detect patterns** that indicate problems (non-linearity, heteroscedasticity, outliers).

Any type of structure in residuals might reveal a **violation of linear regression assumptions**.

Your turn!

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ELSTE Data Science Course

Bivariate analyses

Up to now, the domain of univariate analyses has allowed us to explore the properties of **one variable at the time**. Investigating the relationship between **two variables** is the topic of bivariate analyses. One good visual starting point to identify these is to plot **pairwise relationships** of all the variables in our dataset.

Let's go back to our `df_traces_sub` dataset. We can then plot a pairwise comparison plot using *Seaborn*'s `.pairplot()` function (doc; Listing 1). We use the `hue` argument of the `.pairplot()` function to plot each epoch in a different color:

Listing 1: Pairwise comparison plot using Seaborn